36 Project Management and Execution

By Michael D. Whitt

Topic Highlights

Contracts
- *Constraints*
- *Cost-Plus (CP)*
- *Time and Material (T&M)*
- *Time and Material/Not-To-Exceed (T&M/NTE)*
- *Lump-Sum, or Fixed-Price*
- *Hybrid*

Project Life Cycle
- *Feasibility Study*
- *Definition*
- *System Design*
- *Software Development*
- *Deployment*
- *Support*

Project Management Tools
- *The Scope of Work (SOW)*
- *The Estimate*
- *The Design Schedule*
- *The Status Report*

Project Management Techniques
- *Assessing Project Status*
- *Management of Change (MOC)*

36.1 Introduction

Project: A project is a temporary activity whose purpose is to create a product or service. Temporary projects have a defined beginning and end. Projects usually involve a sequence of tasks with definite starting and ending points. These points are bounded by time, resources, and end results.[1]

An engineering project is a means to an end, with a specific objective. For such a project to exist there must be a perceived need and an expectation that the need can be met with a reasonable investment. The owner must weigh the risks against the rewards and conclude the project is worthwhile. Making that risk/reward assessment is sometimes more of an art than a science. *Every project— particularly an automation project—involves some level of risk.*

1. Cockrell, Gerald W., *Practical Project Management: Learning to Manage the Professional* (ISA, 2001), p. 2.

> *All pilots take chances from time to time, but knowing—not guessing—about what you can risk is often the critical difference between getting away with it and drilling a fifty-foot hole in Mother Earth.*
>
> —Chuck Yeager, 1985[2]

More than in most endeavors, the effects of mishandling risk in an automation project can be catastrophic. Beyond the economic ramifications of a poor estimate, which are bad enough, the potential risk to the operators and the public at large can be extensive. Therefore, a well-conceived process of preliminary evaluation, short- and long-range planning, and project control is necessary. This evaluation begins with a thorough understanding of the issues. Like General Yeager, the key is to know, not to guess.

Proper project management starts with the project manager (PM). What are some attributes of a good PM? A good PM will:

- Understand who his real customer is and gain a thorough understanding of the forces driving the customer/owner (risk-taker) to make the investment in the first place.
- Be the customer's advocate when working within his own organization and be an advocate for his own organization when working with the customer. The PM must, more than any other individual on the project team, live with one foot in each camp.
- Have knowledge relating to the techniques of project management and the technological, logistical, and interpersonal challenges facing him.
- Ensure that each member of the project team has a thorough knowledge of the issues.
- Continually monitor the project's progress, measuring against the agreed-upon parameters.

Whether you are a project manager, or a project team member, a thorough understanding of the principles and concepts discussed here will broaden your avenue to success. The following major topics are discussed in this chapter:

- *Contracts* – What are some of the most common project types?
- *Project Life Cycle* – What is the normal order of things?
- *Project Management Tools* – What is in the project manager's toolbar (i.e. project controls)?
- *Project Execution* – What are some of the key techniques for managing an ongoing project?

36.2 Contracts

Each member of the project team defines *success* from his or her own unique perspective, as viewed through the prism of the project parameters. From the customer's point of view, success is achieved when the desired end is reached within the time allotted and/or the funds allocated. This is likely to be a broader interpretation than that of the service provider, who is also interested in making a profit.

A truly successful project is one in which both the customer and the service provider are satisfied with the outcome. For this to occur, a zone of success must be created that is as large as possible (see Figure 36-1).

A "success triangle" is formed at the point where the goods delivered meet the customer's cost and quality expectations, while still allowing the service provider to make a fair profit and maintain his reputation for good quality work. Staying within this comfort zone is sometimes a bit tricky, and it

2. Yeager, General Chuck and Leo Janos, *Yeager: An Autobiography* (Bantam Books, 1985), p. 84.



Figure 36-1: Success Triangle

really helps to understand the "physics" involved. The physics of a project are defined by its constraints.

36.2.1 Constraints

Time and resources are two parameters that impose limits on the design process (see Figure 36-2). Time, as it relates to a project, can mean either duration (calendar time) or intensity (labor hours). The term *time driven* in this context implies the project is constrained by the calendar; the term *cost driven* implies the project is constrained by cost. Cost is calculated by finding material cost and then measuring the level of intensity per unit of project time (in man-hours) required to design and construct.

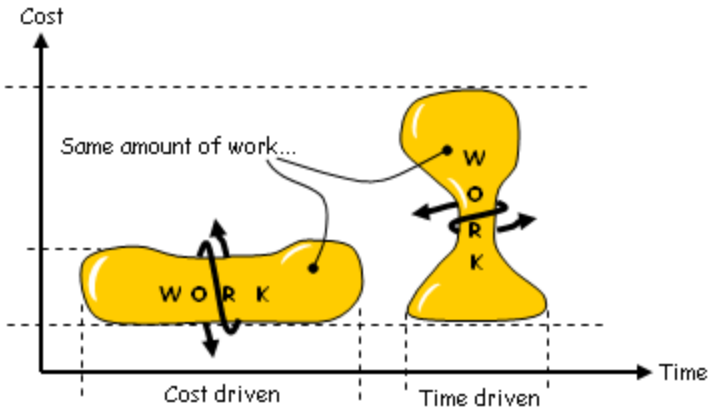


Figure 36-2: Effects of Constraints on Project Structure

The relationship between the customer and the engineer or constructor requires clear definition. If a vendor provides a quote, then that vendor is bound by it, and must provide the materials or services for the price offered in the quote. The same concept applies to the automation service provider (seller). As the seller, the act of submitting a proposal constitutes initiating the contracting process. If the customer (buyer) accepts the proposal, the seller is legally bound to execute according to the contract, and the buyer is legally bound to honor it. Following are some of the most common types of contracts:

- "Cost-Plus"
- "Time and Material" (also, "Time and Material, Not-to-Exceed")
- "Lump Sum" or "Fixed-Fee"

- "Turnkey"
- "Hybrid"

Each contract type strikes a different balance of risk/reward for each participant as noted in the following commentary.

Table 36-1 (below) depicts the relative risk/reward factors of several of the most common types of contracts. Each contract type is analyzed with respect to the project constraint, and rated on a scale of 0 – 4 as follows:

0: None

1: Minimal

2: Moderate

3: Maximum

A risk/reward ratio of 1/3, therefore, would indicate the condition has minimal risk, with maximum reward… a very desirable state, indeed, for the concerned party.

Table 36-1: Risk/Reward Analysis

Contract Type	(Seller)	(Shared)	(Buyer)	Project Constraint
	Profit	**Quality**	**Cost**	
Cost-Plus	0 / 1	0 / *	3 / #	Content
Cost-Plus	0 / 1	2 / *	2 / #	Schedule
T&M	0 / 2	0 / *	3 / #	Content
T&M	0 / 2	2 / *	2 / #	Schedule
T&M, NTE	1 / 2	0 / *	1 / #	Content
T&M, NTE	3 / 2	2 / *	1 / #	Schedule
T&M, NTE	3 / 2	2 / *	1 / #	Cost
T&M, NTE	3 / 2	3 / *	1 / #	Cost & Schedule
Lump Sum	1 / 3	0 / *	3 / #	Content
Lump Sum	3 / 3	2 / *	2 / #	Schedule
Lump Sum	3 / 3	2 / *	0 / #	Cost
Lump Sum	3 / 3	3 / *	0 / #	Cost & Schedule
* High quality is the reward for all parties in all cases… # Low cost is the reward in all cases for the Buyer… ( ) Indicates concerned party: Seller, or Buyer				

The following commentaries discuss each of these major contract types in detail:

36.2.2 Cost-Plus (CP)

A cost-plus contract guarantees a profit for the seller. In return, the buyer may retain control over project content and the seller's method of execution. The parameters defined by the contract are unit rates—not project price. The unit rates can be applied to either the various employees' hourly rates, or to negotiated rates based on employee's classification (e.g., engineer, programmer, designer, clerk,

etc.). These unit rates are valid over the life of the contract, which can extend into the future until the scope of work is satisfied. For the seller, this guarantees a minimal, negotiated profit regardless of project constraint (Profit: no risk, minimal reward). If the seller completes the task below budget, the buyer realizes a windfall. But, if the seller exceeds budget, the buyer must pay. However, if the seller is constrained by schedule, then a sort of budgetary control can be established that can mitigate the risk/reward ratio.

Refer to Table 36-1: If the constraint is content, the seller will eventually get what he wants (Quality: 0, no risk), but with a very real risk of high cost (Cost: 3, high risk). In this scenario, the seller may work indefinitely until the buyer is satisfied with the result. But, if schedule is defined as a constraint, then the risk to quality rises to moderate levels with a corresponding drop in the risk of cost overruns from high to moderate. The seller just has no time to spend an infinite amount of resources. Note that Cost is not a constraint in this format, even though the buyer always has a budget that was approved internally. The buyer's internal budget has no bearing on this type of contract. Though it is in the seller's best interest to know the buyer's target and try to work within it, he is not contractually obligated to do so.

36.2.3 Time and Material (T&M)

The T&M contract and the Cost-Plus contract are very similar. For Cost-Plus, the service provider is reimbursed for his cost, plus profit and expenses. The T&M contract reimburses for cost, plus profit and expenses, plus material and markup. Again, the parameters defined by the contract are unit rates – not project price. Therefore, the only major difference between the two is a higher potential profit for the seller due to his markup on materials (see Table 36-1). Please note that the T&M contract's risk/reward scenarios for Quality and Cost are the same as for the Cost-Plus format. Please refer to the narrative description for Cost-Plus for more insight into this contract type.

Both the T&M and the CP formats are tried-and-true structures that tend toward stable, long-term business relationships between buyer and seller—provided the seller guards the buyer's interests and manages the relationship. The allure of future project work is the carrot that offsets the seller's tendency to become complacent. Good project management techniques are as important for management of these contracts as for any other—perhaps more so, since the need for it is sometimes not readily apparent. See the section on Management of Change (MOC) later in this chapter.

36.2.4 Time and Material/Not-To-Exceed (T&M/NTE)

The CP and T&M scenarios place most of the budgetary risk squarely on the shoulders of the buyer. The "not-to-exceed" stipulation reverses this by adding the constraint of "cost." The seller's potential reward remains moderate, as in the straight T&M format, but his profit risk climbs to very high levels very quickly as constraints are added. In addition, if the seller finishes below budget the buyer receives the windfall—not the seller. High-risk/moderate reward for the seller causes an increase in his intensity and focus, and decreases his desire to work with the buyer to "tweak" the product. Thus, the buyer loses some control over the way the project is executed, and the likelihood of disagreements between the two organizations increases somewhat.

36.2.5 Lump-Sum, or Fixed-Price

The terms lump-sum and fixed-price are interchangeable. In the lump-sum contract, the parameters defined by the contract are a fixed project price for a fixed set of tasks or deliverables. Since the price is negotiated before the services are rendered, the fixed price contract minimizes cost risk for the buyer. Further, the buyer usually signs a contract only after a bidding process in which several potential sellers compete for the contract by providing their lowest bids. The buyer can either accept the lowest bid, or he can analyze them to determine the lowest "reasonable" bid that will save him the most money while letting him retain a sense of comfort that the seller can execute to the terms of the contract.

The profit risk to the seller depends on whether the service or product is "deterministic" or "probabilistic." If selling "widgets," in which little or no research and development is needed, the product is

deterministic, and presumably the seller knows exactly how much resources and funds are required in their production. If making retrofit modifications to an existing facility in which drawings and/or software listings are out of date, for example, then the project is deemed probabilistic, and his or her risk is maximized. One way for the seller to mitigate risk, and also maximize reward, is to add "contingency" to the bid.

Contingency is a factor used to cover normal design development issues that are hard to quantify, and also to cover those things unknown. If the level of uncertainty is low, then the level of contingency can be low. If not, then contingency should be high. The proper amount of contingency reflects the balance between the level of uncertainty that exists, and the level of risk deemed acceptable. Sometimes the seller is willing to forego profit to retain staff, reducing contingency; sometimes the staff is busy, leading the seller to raise it.

Profit, for the seller, is at risk, depending on the scenarios described above. But, the possibility of reward is also high. If the seller manages to execute his or her task below budget, the buyer must pay the full amount of the contract. High risk/high reward.

A fixed-price, or lump-sum, project offers the buyer several benefits, foremost of which is an enhanced ability to allocate resources. The buyer can set aside project funds (plus a safety buffer) with a high level of confidence that additional funds will not be required. This works to the seller's advantage in planning other work. The buyer gains these benefits, but, in comparison to the cost-plus format, loses much control during the execution of the project. To submit fixed-price bids, bidders work at their own expense to clearly define not only the set of deliverables, but the methods they expect to employ to meet the owner's defined scope of work.

A fixed-cost project, if properly managed, can be the most efficient and effective format for both organizations. However, the seller must be ever vigilant in managing scope-creep.[3] Once the buyer accepts a bid, the seller has no obligation to adjust the deliverables or methods if it can be demonstrated that doing so will negatively affect his ability to turn a profit. If the buyer makes a request that is out of bounds with respect to the scope of work, the seller has the right—and obligation—to refuse the request until the buyer approves an engineering change order (See Management of Change). This defensive posture on the seller's part can lead to a fractious relationship with the customer if a previously agreed upon method of change control is not employed.

36.2.6 Hybrid

A particular project may exhibit several of the characteristics in the various scenarios described above. For example, the preliminary engineering (discovery) phase is frequently done on a cost-plus basis because there is simply not enough information available to produce a responsible fixed quote. Subsequent phases such as detail engineering and construction may then be done on a lump-sum basis.

36.3 Project Lifecycle

No matter what style of project, whether lump sum or T&M, most automation projects are similar in the way they are developed and executed. Figure 36-3 depicts the ISA Certified Automation Professional (CAP) program's model for this cycle.

36.3.1 Feasibility Study

Automation projects frequently begin on the production floor. A problem needs to be fixed or a process needs to be streamlined. For the issue to be addressed, someone must isolate the problem and prepare a written description detailing its properties and effects. Some of the areas to consider when describing the effects are as follows:

3. "Scope-creep" describes the case where the seller agrees to perform additional out-of-scope work without amending the contract.

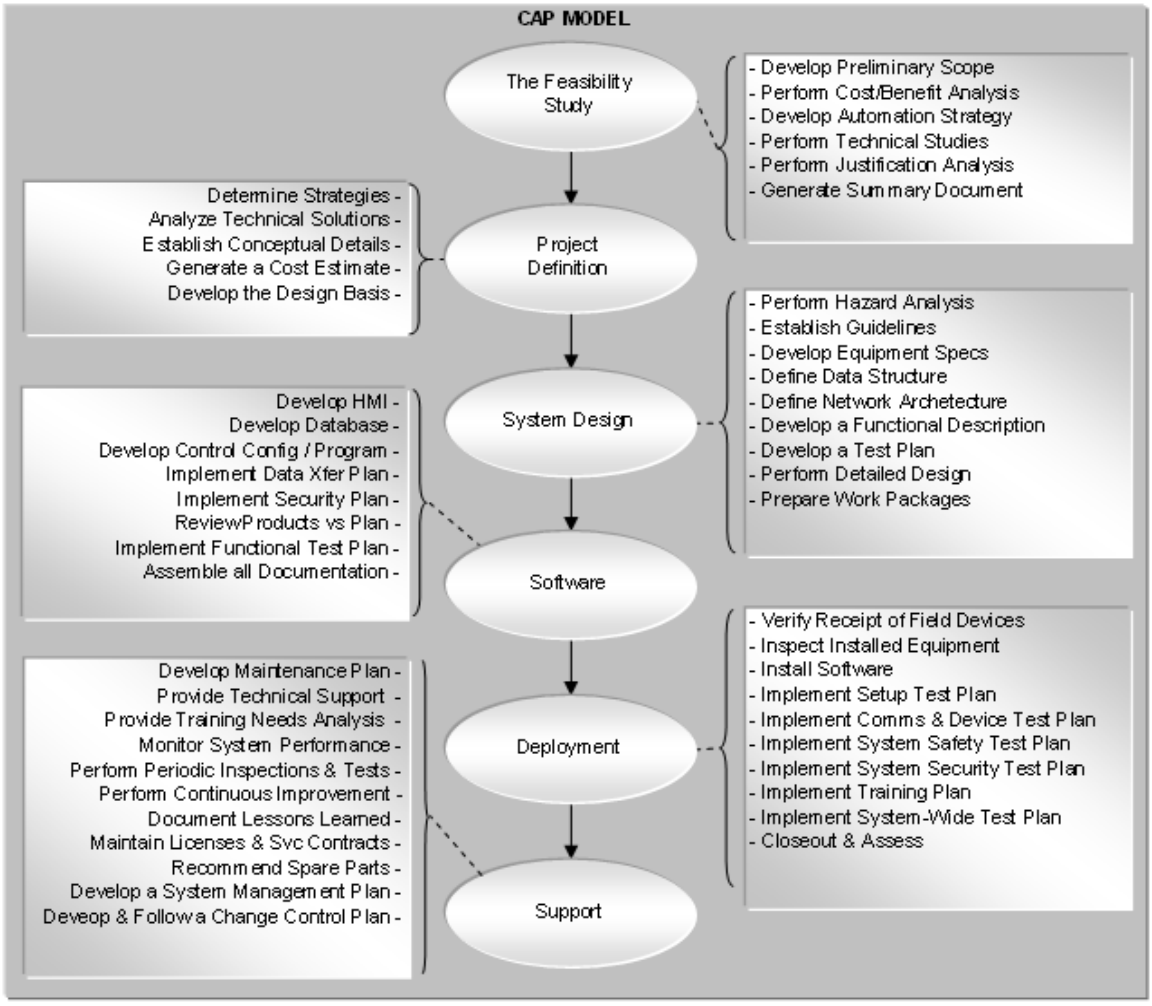


Figure 36-3: ISA-CAP Model for Automation Project Flow

A) Describe the Need
Issues to consider are:

- Personnel safety
- Production rate
- Product quality
- Equipment reliability
- Maintenance
- Operability

B) Identify Possible Solutions

- Analyze the problem
- Estimate the cost

C) Develop a Preliminary Scope of Work
Write a project scope of work that clearly describes the project goals. It should provide enough information to let someone prepare a reasonably accurate estimate with minimal expense. The following information should be discussed in the document:

- Document Lists
- Equipment Lists

- Software/hardware performance specifications
- Service provider performance criteria such as:
  - How should documents be transmitted?
  - What media should be used to prepare the documents? CADD? Manual?
  - What design standards should be adhered to? NFPA? NEC? Internal?
  - What is the desired timetable?
  - What specific deliverables will be expected?
- Approved vendors lists
- Safety concerns
- Security

D) Perform a Cost/Benefit Analysis

The cost/benefit analysis compares the most likely investment cost to the most likely return. The result of this analysis can be expressed in units of calendar-time. For example, if a project costs $10M, and the net profit on the product is expected to be $2M/yr, then the project has a 5-year payback.

E) Develop an Automation Strategy

The automation strategy must take into account operability issues such as plant shutdowns, equipment availability, and the state of the plant's infrastructure. It must accommodate the operations department and the maintenance department's concerns, as well as satisfy technology issues.

F) Perform Technical Studies

Technical studies to prove basic concepts are essential to a successful project. The aim of these studies is to eliminate as many of the variables and unknowns as possible.

G) Perform Justification Analysis

The justification analysis looks at marketplace effects and risk, in addition to cost. This analysis reviews the relative position of the company with respect to its competitors in the marketplace; it forecasts the effects of a successful project, and an unsuccessful project, in the marketplace.

H) Generate a Summary Document

The feasibility study should be published as a document that summarizes the findings of each of the aforementioned activities to support a value assessment of the merits of the project.

36.3.2 Definition

The definition phase of the project identifies customer requirements and completes a high-level analysis of the best way to meet those requirements.

A) Determine Operational Strategies

Key stakeholders should be identified and interviewed to establish the true impetus of the project. What is the need this project is envisioned to fulfill? The service provider should take the time to satisfy himself or herself that the customer has considered all the issues and will be happy with the results of the project if it is executed well. The Law of Unintended Consequences should be considered in this step.

B) Analyze Technical Solutions

Once the true reason for the project is identified, and the operational strategy developed, technical solutions to the problem should be sought. This can entail visits to sites with similar situations, vendor and/or manufacturer interviews, and sometimes test-bed evaluations or pilot projects.

C) Establish Conceptual Details

Several sets of documents are produced during the conceptual stages of the project. These documents are referred to as "upper-tier" documents. They describe the conceptual details to develop a design basis.

D) Generate a Cost Estimate

The conceptual details developed above provide a baseline to which the design may be applied. A detailed Work Breakdown Structure[4] (WBS)-based estimate can now be produced. Once the "first-draft" of the estimate is done, a review cycle should be used to look for cost-reduction characteristics such as economy of scale, duplication of work, etc.

E) Develop the Design Basis

The Design Basis is a detailed plan of execution for the System Design Phase. A complete list of deliverables is developed that is based on the research done in the Definition Phase.

36.3.3 System Design

Steps to the "System Design" phase of the project are:

A) Perform Hazard Analysis (HAZOP)[5]

After the design basis is defined to a fairly high degree, but before the detail design has begun, the envisioned system should be analyzed for operability and safety.

B) Establish Guidelines

Establish standards, templates, and guidelines as applied to the automation system using the information gathered in the definition stage and considering human-factor effects to satisfy customer design criteria and preferences.

C) Develop Equipment Specifications and Instrument Data Sheets

Create detailed equipment specifications and instrument data sheets based on vendor selection criteria, characteristics and conditions of the physical environment, regulations, and performance requirements to purchase long-lead-time equipment and support system design and development.

D) Define the Data Structure Layout and Data Flow Model

Define the data structure layout and data flow model, considering the volume and type of data involved to provide specifications for hardware selection and software development.

E) Select the Physical Communication Media, Network Architecture, and Protocols

Select the physical communication media, network architecture, and protocols based on data requirements to complete system design and support system development.

F) Develop a Functional Description of the Automation Solution

Develop a functional description of the automation solution (e.g., control scheme, alarms, HMI, reports) using rules established in the definition stage to guide development and programming.

G) Develop a Test Plan

Design the test plan using chosen methodologies to execute appropriate testing relative to functional requirements.

H) Perform Detail Design

Perform the detailed design for the project by converting the engineering and system design into purchase requisitions, drawings, panel designs, and installation details consistent with the specification and functional descriptions to provide detailed information for development and deployment.

I) Prepare Construction Work Packages

Prepare comprehensive construction work packages by organizing the detailed design information and documents to release project for construction.

4. Work Breakdown Structure: A task-based method of organizing a project, usually sub-grouped by process area or physical location.
5. HAZOP: Hazard and Operability Study. See *Process Safety and Safety Instrumented Systems* chapter.

J) Procurement
Several procurement activities occur during the latter part of the design phase. Delivery lead times must be considered with respect to timing of orders.

36.3.4 Software Development

If the system design phase is properly executed, the software development phase of the project should be simply a matter of executing to the plan. In most cases, the content being developed in the areas described below are defined in the plans produced previously. Following are some of the steps involved in developing software:

A) Develop the Human-Machine Interface (HMI) System using:
- The Alarm Grouping and Annunciation Plan
- The Alarm and Alarm Setpoint List
- The *Human-Machine Interface (HMI)* Screen Hierarchy and Navigation Plan
- The HMI Color Scheme and Animation Plan
- The Operational Security Plan

B) Develop Database and Reporting Functions using:
- The Data Historian List and Archival Plan
- The Data Backup and Restore Plan
- The Report and Report Scheduling Plan

C) Develop the Control System Configuration and/or Program using: [6]
- Logic Diagrams
- Device Control Detail Sheets (DCDS)
- Process Control Detail Sheets (PCDS)
- Sequence Control Detail Sheets (SCDS)
- Recipes
- Interlock Lists and Alarm/Trip Setpoint Lists

D) Implement the Data Transfer Plan
Implement data transfer methodology that maximizes throughput and ensures data integrity using communication protocols and specifications to assure efficiency and reliability.

E) Implement the Operational Security and Data Integrity Plan
Implement security methodology in accordance with stakeholder requirements to mitigate loss and risk.

F) Perform a Scope Compliance Review
Review configuration and programming using defined practices to establish compliance with functional requirements.

G) Develop and Implement the Functional (or Factory) Acceptance Test Plan (FAT)
Test the automation system using the test plan to determine compliance with functional requirements.

H) Assemble Documentation, and Prepare for Turnover
Assemble all required documentation and user manuals created during the development process to transfer essential knowledge to customers and end users.

6. Many of the deliverables listed in this section are described fully in *Successful Instrumentation and Control Systems Design* by Michael Whitt.

36.3.5 Deployment

The "deployment" phase, or construction phase, is the phase in which the design is implemented. The deployment phase frequently overlaps the design phase activities by some margin, beginning construction as WBS area designs are completed. During this time of overlap, the needs of the constructor ascend in importance over the needs of design. Work stoppages should be avoided at all costs, even to the point of interrupting ongoing design activities to concentrate on a phase-three problem.

Upon arrival at the site, the automation professional should begin the processes described below.

A) Verify Field Device Status

Perform receipt verification of all field devices by comparing vendor records against design specifications to ensure devices are as specified.

B) Inspect Installed Equipment

Perform physical inspection of installed equipment against construction drawings to ensure installation in accordance with design drawings and specifications. Construction activity status should be captured in a report similar to the Startup Readiness Report, shown below.

Startup Readiness Report

	Installation				Checkout		
Tag Number	Mounted?	Wired?	Tubed?	Rung Out?	Local	Remote	Summary
PT-10	1	1	1	1	1	1	100.0%
PT-11	1	1	1	1	NA		80.0%
PT-12	1		1				33.3%
LT-15							0.0%
PCV-10	1	1					33.3%
HV-11							0.0%

Excel readily lends itself to this type of use. The Summary field can be used to help develop a completion percentage if desired. In this case, the summary assumes each category is weighted equally. The numbers are summed and then simply divided by the number of categories. PT-10 has six categories and PT-11 has five. The equation "=SUM(C5:H5)/6" is loaded into the summary cell for PT-10, for example.

Figure 36-4: Startup Readiness Report

Using the instrument database and I/O list as a basis, a Startup Readiness Report can be produced that will make this an organized, verifiable process.

C) Install Hardware and Software

Install configuration and programs by loading them into the target devices to prepare for testing.

D) Perform Preliminary System Checks; Troubleshoot and Correct Faults

Solve unforeseen problems identified during installation using troubleshooting skills to correct deficiencies.

E) Perform the Site Acceptance Test for the Control Software

Test configuration and programming in accordance with the design documents by executing the test plan to verify the system operates as specified.

F) Perform the Site Acceptance Test for Communications and Field Devices
Test communication systems and field devices in accordance with design specifications to ensure proper operation. This is sometimes referred to as the "checkout," or "bump and stroke" phase, each field device should be exercised to demonstrate proper operation.

G) Perform the Site Safety Test
Test all safety elements and systems by executing test plans to ensure safety functions operate as designed.

H) Perform the Site Security Test
Test all security features by executing test plans to ensure security functions operate as designed.

I) Execute the Training Plan
Provide initial training for facility personnel in system operation and maintenance through classroom and hands-on training to ensure proper use of the system.

J) Perform the Site System Integrity Test (i.e., Startup & Commissioning)
Execute system-level tests in accordance with the test plan to ensure the entire system functions as designed. A successful test at this stage constitutes the system startup. The system is generally considered to be "commissioned" upon conclusion of this step. Startup should be well organized and sequential. A formal startup procedure is recommended.

K) Closeout & Assess
The final step in the deployment phase is the closeout step. This is the step in which:

- Project documentation is finalized to reflect as-built conditions (thus documenting changes that accrued during the deployment process). Redline markups that occurred due to construction adjustments should be transcribed into the original design package.
- Project documents are turned over to the end user.
- Final billing issues are resolved.
- A post-mortem is held to evaluate and discuss lessons learned, and to recognize high achievement.
- A project celebration is held to commemorate conclusion of the project. Recognizing the staff for their work is of great importance.

36.3.6 Support
The "support" phase of the project should be very important to the automation professional. In many cases, this is where the next big project is derived.

A) Develop and Implement a System Performance Monitoring Plan
Verify system performance and records periodically using established procedures to ensure compliance with standards, regulations, and best practices.

B) Develop and Implement a Long-Term Technical Support Plan
Provide technical support for facility personnel by applying system expertise to maximize system availability.

C) Develop and Implement a Long-Term Training Plan
Perform training needs analysis periodically for facility personnel using skill assessments to establish objectives for the training program.

D) Provide Periodic Training for Facility Personnel
Provide training for facility personnel by addressing identified objectives to ensure the skill level of personnel is adequate for the technology and products used in the system.

E) Monitor System Performance
Monitor performance using software and hardware diagnostic tools to support early detection of potential problems.

F) Perform Periodic Inspections and Tests to Re-certify the System
Perform periodic inspections and tests in accordance with written standards and procedures to verify system or component performance against requirements.

G) Develop and Implement a Continuous Improvement Plan
Perform continuous improvement by working with facility personnel to increase capacity, reliability, and/or efficiency.

H) Document Lessons Learned
Document lessons learned by reviewing the project with all stakeholders to improve future projects.

I) Develop and Implement a License and Service Contract Maintenance Plan
Maintain licenses, updates, and service contracts for software and equipment by reviewing both internal and external options to meet expectations for capability and availability.

J) Provide a Recommended Spare Parts List
Determine the need for spare parts, based on an assessment of installed base and probability of failure, to maximize system availability and minimize cost.

K) Develop a System Management Plan
Provide a system management plan by performing preventive maintenance, implementing backups, and designing recovery plans to avoid and recover from system failures.

L) Develop and Implement a Change Control Plan
Follow a process for authorization and implementation of changes in accordance with established standards or practices to safeguard system and documentation integrity.

36.4 Project Management Tools

A well-conceived, well-managed project is one that has been given a chance to succeed. A set of tools have been developed to help the project team stay on track. If properly developed and used, these tools will help define the tasks to be performed, facilitate accurate progress reporting, and provide early warning of potential cost overruns and/or scheduling conflicts. These tools include the scope of work, the project estimate, the project schedule, and various project reports.

36.4.1 The Scope of Work (SOW)

A scope of work is simply a description of tasks that need to be accomplished to achieve the desired end. It is the result of a cycle of investigation intended to eliminate or reduce the number of assumptions that need to be made later. Thus, an SOW should list each major task for each process area that falls under the scope of the project. Each task should then be evaluated from the perspective of each discipline. All execution statements should begin with verbs to convey the type of work being done.

Each task should have a unique number called a work breakdown structure (WBS) number. An estimate and material cost will be derived for this item and stored as part of the scope of work.

36.4.2 The Estimate

The estimate is an under-utilized tool for the project manager. All too often, a project is estimated using a set of criteria that have no bearing on how the project will be executed, yielding a throw-away document that has little or no use beyond generation of a price. The price generated might be accurate, but an opportunity has been lost to provide meaningful structure to the project. A better way is to build an estimate formatted to reflect the execution plan in the scope of work.

The relationship between the estimate and the scope of work results in a success envelope that is either big enough or too tight. Scope and schedule define the "mechanics" of the project. The estimate defines the "physics" of the situation, setting limits that affect the way in which the project will be executed. These physical measures are the primary tools of the project manager when tracking the project on the strategic level and of the design supervisor managing the work on the tactical level. The scope of work defines the work to be done. The material cost estimate and labor cost estimate define the parameters within which the work will be done.

There are three main types of estimates: budget, bid, and definitive. Following is a brief description of each:

- The *budgetary cost and labor estimate* is produced by a group intimate with the site and process. This group may or may not execute the remainder of the project. The purpose of this type of estimate is primarily to obtain initial funding. It is typically "quick and dirty" and is expected to be rather inaccurate. In fact, an error margin of ±30% is acceptable for this type of estimate. Prior to this estimate, a formal scope of work has probably not been done and, quite possibly, the project specification has not been finalized. This type of estimate is generally unfunded by the customer, or at least under-funded.

- The *bid material cost and labor estimate* is produced by the various engineering bidders vying for the work. Again, this estimate is typically unfunded by the customer.

- The *definitive material cost and labor estimate* is prepared by the engineering firm that was awarded the contract based on their bid. The customer typically includes this as a part of the project, so it probably is fully funded. Once the contract has been awarded and any secrecy agreement issues have been settled, the engineering contractor is given full access to the information developed by the customer during the internal evaluation process. The engineering firm then does some research to validate the basic assumptions and develops a finalized scope of work. This information sometimes alters the picture significantly, and the engineering contractor is given an opportunity to adapt the estimate and schedule to the new information. Of course, this re-estimating process is bypassed if full disclosure was made during the bid process. In that case, the bid estimate becomes the definitive estimate.

The estimate becomes the baseline document for project management, providing a yardstick to which performance will eventually be measured. It is based on the scope of work and should reflect the work scope on a task-by-task basis, wherever possible. Many estimating tools are available. Whichever is chosen, you should be able to produce estimates that are:

- *accurate*, by identifying each task relating to the scope of work and then quantifying the labor and expense of its execution.

- *timely*, by being repeatable.

- *verifiable*, by adding references, notes and amplifications, and by making the calculations available for dissection.

- *meaningful*, by relating all the deliverables to hard data.

- *adaptable*, by being easy to modify to accommodate "what-if" scenarios.

36.4.3 The Design Schedule

A schedule can be built from the project estimate, provided the estimate has enough detail and is task-oriented. Each task should be given a unique WBS identifier. Linked relationships should be identified and analyzed. In most cases, these relationships may be defined as follows:

- *Start-to-start*: Task B can't start until Task A starts. This relationship is used when an outside trigger initiates a scheduled chain of events. For example, receipt of a specific material shipment might trigger several tasks at the same time.
- *Finish-to-start*: Task B can't start until Task A finishes.
- *Start-to-finish*: Task A can't finish until Task B starts.
- *Finish-to-finish*: Task B can't finish until Task A finishes.

After the relationships are defined and the resources have been assigned to each task, the overall project duration may be estimated. The customer may have a time frame in mind already, which constrains the schedule to a particular set of milestones.[7] When the schedule is compared to this time line, resources may need to be modified to lengthen or shorten the schedule.

36.4.4 The Status Report

Periodically, the design team will be asked to provide some feedback as to their execution status. This feedback will answer questions such as the following: Did you start the project on time? How far along are you? Which tasks have you started, and when did you start them? Will you finish the project on time and within budget?

Using the WBS milestone schedule is a way to subdivide the available man hours and/or costs. This effectively breaks the budget into smaller, easier to manage sections that can be reported against individually. Data should be collected for each task (WBS item) and then plowed back into the overall project schedule by the PM. Status reporting typically includes updating the following parameters:

- Start date.
- Finish date.
- Percent complete.
- Man-hours estimate to complete (Manhrs ETC).

Figure 36-5 shows a status data collection form that has 10 WBS groupings. Each grouping is subdivided by six subtasks, which, in this case, equates to the six phases of a project (CAP Model), discussed previously. Of the 10, only WBS-T01 and WBS-T02 have been started. WBS-T01 covers the Railcar Unloading area of the plan, and has 250-manhours allocated. Ideally, the estimate would have been done in this format, with each subtask being estimated on its own merit, and the hours rolling up into the main task equaling 250. The Subtask Weight must equal 100%.

Total manhours, subtask weight, and subtask manhours values were loaded at the beginning of the project, forming the baseline data to which all updates are compared. The data to be updated are the project timeline dates and the Manhrs ETC. The scheduling team plows the date information back into the schedule, and the project manager plows the manhour data back into the budget for analysis (to be discussed later). Notice that the ETC for WBS-T01 shows 188 manhours to complete, with 250 available. From this, it might be inferred the design staff is 188/250 = 75% complete. This is incorrect, as we shall see.

7. Milestone – A schedule item pegged to a particular date. These usually represent targeted points in the engineering process, such as "issue drawings."

Project Status Reporting Structure - by WBS Area									
WBS Item#	Description	Project Timeline Start	Finish	Float	Total Mhrs	Subtask Weight	Subtask Manhrs	% Comp	Manhrs ETC
T01	Railcar Unloading	04/03/05	03/01/07		250.0				188.0
T01 -1	Feasibility Study	04/03/05	04/20/05	5d		10.0%	25.0	100%	-
T01 -2	Definition	05/01/05	06/01/05	15d		10.0%	25.0	95%	8.0
T01 -3	System Design	06/15/05	07/15/05	15d		25.0%	62.5	25%	40.0
T01 -4	Software	08/01/05	09/01/05	15d		20.0%	50.0	10%	60.0
T01 -5	Deployment	12/15/05	02/15/06	15d		25.0%	62.5	0%	56.0
T01 -6	Support	03/01/06	03/01/07	na		10.0%	25.0	0%	24.0
T02	Bulk Storage	04/03/05	03/01/07		400.0				276.0
T02 -1	Feasibility Study	04/03/05	04/20/05	5d		10.0%	40.0	100%	16.0
T02 -2	Definition	05/01/05	06/01/05	15d		10.0%	40.0	95%	32.0
T02 -3	System Design	06/15/05	07/15/05	15d		25.0%	100.0	25%	16.0
T02 -4	Software	08/01/05	09/01/05	15d		20.0%	80.0	10%	72.0
T02 -5	Deployment	12/15/05	02/15/06	15d		25.0%	100.0	0%	100.0
T02 -6	Support	03/01/06	03/01/07	na		10.0%	40.0	0%	40.0
T03	River Water	04/03/05	03/01/07		80.0				80.0
T04	Fire Protection	04/03/05	03/01/07		400.0				376.0
T05	Material Handling	04/03/05	03/01/07		1,000.0				980.0
T06	Crush & Slurry	04/03/05	03/01/07		1,500.0				1,490.0
T07	Convey	04/03/05	03/01/07		400.0				400.0
T08	Truck Loading	04/03/05	03/01/07		80.0				80.0
T09	System Services	04/03/05	03/01/07		80.0				80.0
T10	Miscellaneous	04/03/05	03/01/07		80.0				80.0
	Project Summary:	04/03/05	03/01/07		4,270.0				4,030.0

Figure 36-5: Reporting Project Status

36.5 Project Management Techniques

36.5.1 Assessing Project Status

The project status update data that was collected in Section 36.4.4 can be used by the project manager to develop additional data that will help forecast the likelihood of success for the project (See Figure 36-6). Some of the additional information needed will be:

- *Earned hours = Estimated percent complete X budgeted man hours.*
- *Actual hours:* The number of hours actually expended to date, as reported by the timesheet system.
- *Estimate to complete (ETC):* Entered by the design team to reflect the amount of work remaining.
- *Estimate at completion (EAC) = Manhrs actual + manhrs ETC;* Indicates the expected final manhrs.
- *Apparent percent complete = Manhrs actual/allocated manhrs; m*easures actual vs. initial expectations.
- *Actual percent complete = Manhrs actual/manhrs EAC; m*easures actual vs. final expectations.
- *Efficiency rating = Actual percent complete/apparent percent complete.*

Project Status Reporting Structure - by WBS Area

WBS Item#	Description	Project Timeline Start	Finish	Float	Budget Mhrs	Subtask Weight	Allocated Manhrs	Est. %Comp	Manhrs Earned	Manhrs Actual	Manhrs ETC	Manhrs EAC	Apparent % Comp	Actual %Comp	Efficiency Rating
T01	Railcar Unloading	04/03/05	03/01/07		250.0			42%	69.4	106.0	188.0	294.0	42%	36%	0.85
T01 -1	Feasibility Study	04/03/05	04/20/05	5d		10.0%	25.0	100%	25.0	32.0	-	32.0	128%	100%	0.78
T01 -2	Definition	05/01/05	06/01/05	15d		10.0%	25.0	95%	23.8	40.0	8.0	48.0	160%	83%	0.52
T01 -3	System Design	06/15/05	07/15/05	15d		25.0%	62.5	25%	15.6	32.0	40.0	72.0	51%	44%	0.87
T01 -4	Software	08/01/05	09/01/05	15d		20.0%	50.0	10%	5.0	2.0	60.0	62.0	4%	3%	0.81
T01 -5	Deployment	12/15/05	02/15/06	15d		25.0%	62.5	0%	-	-	56.0	56.0	0%	0%	-
T01 -6	Support	03/01/06	03/01/07	na		10.0%	25.0	0%	-	-	24.0	24.0	0%	0%	-
T02	Bulk Storage	04/03/05	03/01/07		400.0			18%	111.0	72.0	276.0	348.0	18%	21%	1.15
T02 -1	Feasibility Study	04/03/05	04/20/05	5d		10.0%	40.0	100%	40.0	16.0	16.0	32.0	40%	50%	1.25
T02 -2	Definition	05/01/05	06/01/05	15d		10.0%	40.0	95%	38.0	32.0	32.0	64.0	80%	50%	0.63
T02 -3	System Design	06/15/05	07/15/05	15d		25.0%	100.0	25%	25.0	16.0	16.0	32.0	16%	50%	3.13
T02 -4	Software	08/01/05	09/01/05	15d		20.0%	80.0	10%	8.0	8.0	72.0	80.0	10%	10%	1.00
T02 -5	Deployment	12/15/05	02/15/06	15d		25.0%	100.0	0%	-	-	100.0	100.0	0%	0%	-
T02 -6	Support	03/01/06	03/01/07	na		10.0%	40.0	0%	-	-	40.0	40.0	0%	0%	-
T03	River Water	04/03/05	03/01/07		80.0			0%	-	-	80.0	80.0	0%	0%	-
T04	Fire Protection	04/03/05	03/01/07		400.0			0%	-	-	376.0	376.0	0%	0%	-
T05	Material Handling	04/03/05	03/01/07		1,000.0			0%	-	-	980.0	980.0	0%	0%	-
T06	Crush & Slurry	04/03/05	03/01/07		1,500.0			0%	-	-	1,490.0	1,490.0	0%	0%	-
T07	Convey	04/03/05	03/01/07		400.0			0%	-	-	400.0	400.0	0%	0%	-
T08	Truck Loading	04/03/05	03/01/07		80.0			0%	-	-	80.0	80.0	0%	0%	-
T09	System Services	04/03/05	03/01/07		80.0			0%	-	-	80.0	80.0	0%	0%	-
T10	Miscellaneous	04/03/05	03/01/07		80.0			0%	-	-	80.0	80.0	0%	0%	-
	Project Summary:	04/03/05	03/01/07		4,270.0				180.4	178.0	4,030.0	4,208.0	4%	4%	1.01

Note: Efficiency rating is calculated by dividing "Apparent % Comp" by "Actual % Comp".

Figure 36-6: Analyzing Project Status

Note the differences between the estimated, apparent, and actual completion percentages. In the case of WBS-T01-2, the design staff thought they were 95% complete (estimated). But, they had already exceeded the original budget by 60% (apparent), and should have been done a long time ago. From the standpoint of the EAC, the one that counts, they are really at 83% (actual).

36.5.2 Management of Change (MOC)

Every project, regardless of the type of contract, has both implicit and explicit sets of expectations. This is because it is impossible to spell out every permutation in a contract. Project parameters as noted in the contract (explicit expectations) can be met, and yet the project can still fail due to poor relationships, misunderstandings, bad perceptions, and the like (implicit expectations).

Thinking back to the Success Triangle (Figure 36-1), the buyer's focus is on quality and cost; the seller's focus is on quality and profit. Few projects get off the ground if either party knows that the triangle is either too small, or non-existent. However, projects still fail. So, in those cases there must be a disconnect between what is suspected at the beginning, and what is known at the end.

The key to this mystery is both parties' ability to manage change. The type of contract has a bearing on the amount of energy that is put into the MOC process, but MOC needs to be addressed in some fashion across the board. In fact, in the most benign case of the cost-plus contract, the buyer still has an expectation of ultimate price. If the scope of work is allowed to expand unchecked over a long period of time, no budgetary quote is safe and the likelihood of the buyer's remaining happy with the seller, regardless of the seller's abilities, is remote.

Therefore, it is important that the management of change (MOC) topic be discussed openly and frankly at the project's inception. A procedure for processing change notices should be approved by all parties, and documented thoroughly—even to the point of amending the contract if necessary.

To manage change, a good baseline should be established. This baseline is the design basis. The design basis includes not only the list of deliverables, but also the project execution plan (PEP). The PEP incorporates the scope of work, and details methods and assumptions. Every member of the project team should be cognizant of the design basis, and should continually compare the tasks being done to

that document. As changes are discovered, a change notice should be generated to be passed through the management team for review and approval.

36.6 References

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About the Author

Michael D. Whitt's experience includes 15-years with Raytheon Engineers and Constructors as an I&C Design Supervisor; five years with Mesa Associates, Inc., as Industrial Controls Department Manager responsible for Instrumentation Engineering & Design, Systems Integration, and Panel Fabrication groups. He is the author of *Successful Instrumentation and Control Systems Design* (ISA, 2004). He is a native of Asheville, N.C.